Projective measurements:

A projective measurement is given by an observable M, a hermitian $(M^\dagger = M)$ on the state space of the system.

$$M = \sum_m m P_m$$

P_m is the projector onto the eigenspace of M with eigenvalue m.

$$P_m = |m\rangle\langle m|$$

$$M_0 = |0\rangle\langle 0| \qquad M_1 = |1\rangle\langle 1|$$

$p(m) = \langle\psi|P_m|\psi\rangle = |\langle m|\psi\rangle|^2 =$ probability of measuring eigenvalue m

$M_m = |m\rangle\langle m|$

measurement operator with an index of the outcome

$$\sum_m M_m^\dagger M_m = I.$$

$$M_0^\dagger M_0 + M_1^\dagger M_1 = I$$

generalization (single qubit)

Classical probability theory

$$X : E(X) = \sum_x x\,p(x) = \int x\,p(x)\,dx = \bar{x}$$

$$p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle = \langle\psi|P_m|\psi\rangle.$$

$$E(M) = \sum_m m P_m.$$

$E(M) =$ expectation value in a given state $|\psi\rangle$

$$= \langle\psi|M|\psi\rangle$$

$$E(M) = \sum_m m\langle\psi|P_m|\psi\rangle$$

$= m\,p(m)$ mimics. $E(X) = \sum_x x\,p(x)$

Standard Deviation

$$[\Delta(M)]^2 = \langle M^2\rangle - \langle M\rangle^2 = \langle (M - \langle M\rangle)^2\rangle$$

$$\Delta M = \sqrt{\langle M^2\rangle - \langle M\rangle^2}$$

$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$

Suppose we have a state $|\psi\rangle$, M.

$M|\psi\rangle = m|\psi\rangle$ unit vector

$$\langle M\rangle = \langle\psi|M|\psi\rangle = m\langle\psi|\psi\rangle = m$$

$$\langle M^2\rangle = \langle\psi|M^2|\psi\rangle = \langle\psi|M^2|\psi\rangle = m^2\langle\psi|\psi\rangle = m^2$$

$$(\Delta M)^2 = \langle M^2\rangle - \langle M\rangle^2 = m^2 - (m)^2 = 0$$

For general operators C, D

$$\Delta C\,\Delta D \geq \frac{|\langle\psi|[C,D]|\psi\rangle|^2}{2}$$

Heisenberg uncertainty principle.

$$\Delta x\,\Delta p \geq \frac{\hbar}{2}$$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = |0\rangle\langle 0| - |1\rangle\langle 1|$$

$+1 \quad |0\rangle$

$-1 \quad |1\rangle$

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$\langle\psi|Z|\psi\rangle = \frac{1}{2} - \frac{1}{2} = 0$$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = |+\rangle\langle +| - |-\rangle\langle -|$$

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$X|0\rangle = |1\rangle \qquad X|1\rangle = |0\rangle$$

$$|0\rangle = |+\rangle + |-\rangle \qquad |1\rangle = |+\rangle - |-\rangle$$

$$\langle 0|X|0\rangle = 0$$

$$\langle 0|X^2|0\rangle = 1$$

$$X^2 = I.$$

$$\langle X^2\rangle - \langle X\rangle^2 = 1 - 0 = 1$$

Since $|0\rangle$ is not a eigenstate of X $\langle\psi|X|\psi\rangle$ but the S.D is 1 which tell us that this state is a mixture of eigenstates of X.

POVM: Positive operator valued measurements.

Let M_m (measurement operator) on a quantum system in state $|\psi\rangle$

$$p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle$$

$$E_m = M_m^\dagger M_m$$

$$p(m) = \langle\psi|E_m|\psi\rangle.$$

$$\sum_m p(m) = 1$$

$$\Rightarrow \boxed{\sum_m E_m = I.}$$

The set of operators $\{E_m\}$ define a POVM which is sufficient to determine the probabilities of all possible outcomes.

$\{E_m\}$ is called a POVM

Projective measurements are an example of POVM.

Let Alice and Bob

Alice prepares two states $|\psi_1\rangle = |0\rangle$

$$|\psi_2\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Alice gives these states to Bob, he knowns that he got two states $|\psi_1\rangle, |\psi_2\rangle$ but he doesn't know which one is $|0\rangle$ and which one is $\frac{|0\rangle + |1\rangle}{\sqrt{2}}$

Bob defines a POVM.

$$E_1 = \frac{\sqrt{2}}{1+\sqrt{2}}|1\rangle\langle 1|$$

$$E_2 = \frac{\sqrt{2}}{1+\sqrt{2}}\frac{(|0\rangle - |1\rangle)}{\sqrt{2}}\left(\frac{\langle 0| - \langle 1|}{\sqrt{2}}\right)$$

$$E_3 = I - E_2 - E_3 \Rightarrow \boxed{E_1 + E_2 + E_3 = I}$$

He picks a state call's it $|\varphi\rangle$

$|\varphi\rangle$

$\langle\varphi|E_1|\varphi\rangle = 0 \qquad |\varphi\rangle = |\psi_1\rangle = |0\rangle$

$\langle\varphi|E_1|\varphi\rangle \neq 0 \qquad |\varphi\rangle \neq |\psi_1\rangle$

$\langle\varphi|E_2|\varphi\rangle = 0 \qquad |\varphi\rangle = |\psi_2\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$

since $E_2 = \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)\left(\frac{\langle 0| - \langle 1|}{\sqrt{2}}\right)$

$\langle\varphi|E_2|\varphi\rangle \neq 0$

$\Rightarrow |\varphi\rangle \neq |\psi_2\rangle$

$$\frac{\langle 0| - \langle 1|}{\sqrt{2}} = |\alpha\rangle \qquad \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |\beta\rangle \qquad \langle\alpha|\beta\rangle = 0$$

But let's say he uses E_3: then we can nothing about the state!

Bob never makes mistake in inference.

A point to be noted.

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \qquad |\psi'\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

in terms of probabilities of measurement, they look identical.

$$p(|0\rangle) = \frac{1}{2} \qquad p(|1\rangle) = \frac{1}{2}$$

But there is a relative phase. is also irrelevant for probabilities.

$$|\psi'\rangle = \frac{|0\rangle + e^{i\pi}|1\rangle}{\sqrt{2}} \qquad \langle\psi'|\psi\rangle = 0$$

$$|\psi'\rangle = e^{i\pi/4}\frac{(|0\rangle + |1\rangle)}{\sqrt{2}}$$

global phase: because it will disappear in probabilities.

In a multi-qubit system under unitary evolution, relative phases will interfere.

Entanglement:

Let's consider $|\psi\rangle = \frac{|0\rangle\otimes|0\rangle + |1\rangle\otimes|1\rangle}{\sqrt{2}}$

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$\exists$ no single qubit states $|a\rangle, |b\rangle$ such that $|\psi\rangle = |a\rangle\otimes|b\rangle$

A state of the composite system which can't be written as a tensor product (product state) is called an entangled state.

$$|a\rangle = \alpha|0\rangle + \beta|1\rangle \qquad |\alpha|^2 + |\beta|^2 = 1$$

$$|b\rangle = \gamma|0\rangle + \lambda|1\rangle \qquad |\gamma|^2 + |\lambda|^2 = 1$$

such that

$$\boxed{|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}} = (\alpha|0\rangle + \beta|1\rangle)\otimes(\gamma|0\rangle + \lambda|1\rangle)$$

no values of $\alpha, \beta, \gamma, \lambda$ can satisfy this equation.

Example of entangled states as a mean of sending information in a secure manner.

Alice and Bob

Alice wants to send two bits of information to Bob and they share an entangled state.

They both had their own qubit. They let them interact and prepare the two-qubit system in an entangled state.

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

if Alice wants to send

00 : she does nothing to her qubit and send it to Bob

01 : She applies phase flip Z operator to her qubit and then send

10 : She applies X gate (NOT gate) to her qubit and send.

11 : she applies iY to her qubit and send it to Bob.

00 : $|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$ (1)

01 : $Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle$

$\frac{|00\rangle - |11\rangle}{\sqrt{2}} = |\psi\rangle$ (2)

10 : $X|0\rangle = |1\rangle \quad X|1\rangle = 0$

$\frac{|01\rangle + |10\rangle}{\sqrt{2}} = |\psi\rangle$ (3)

11 : $iY|0\rangle = -i|1\rangle \quad iY|1\rangle = +i|0\rangle$

$\frac{|01\rangle - |10\rangle}{\sqrt{2}} = |\psi\rangle$ (4)

Four orthogonal states: Bell states

$$E_1 = \alpha\left(\frac{|00\rangle + |11\rangle}{\sqrt{2}}\right)\left(\frac{\langle 00| + \langle 11|}{2}\right)$$

$$E_2 = \beta\left(\frac{|00\rangle - |11\rangle}{\sqrt{2}}\right)\left(\frac{\langle 00| - \langle 11|}{\sqrt{2}}\right)$$

$E_3, E_4 \qquad E_1 + E_2 + E_3 + E_4 = I$

POVM on the two qubit system

With a single qubit, Alice sent two bits of classical information.